Nonextendibility of symmetric one-dependent four-color laws to a branched ray

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Abstract

Let a graph consist of a root, two leaves adjacent to it, and an infinite ray starting at the root. We prove that this graph admits no color-symmetric, one-dependent proper four-coloring whose law on every ordered simple path agrees with a common stationary, reflection-invariant line law. The line law is arbitrary. An infinite color-avoidance recurrence bounds every specified one-step extension by half the probability of its prefix. Together with finite marginal identities, this gives an exact rational contradiction. The finite calculation is computer-assisted; a complete certificate and a verifier using only rational arithmetic are supplied.

1 The problem

One-dependence requires restrictions to vertex sets at distance greater than one to be independent. It is compatible with proper coloring on the integer line: Holroyd and Liggett constructed a stationary one-dependent four-coloring whose law is invariant under color permutations and reflection [1, Theorem 4]. A branch imposes additional consistency conditions on the same path probabilities.

Liggett and Tang proved a nonextension result for the prescribed Holroyd–Liggett line law [2, Proposition 1.4]. The paragraph following that proposition distinguishes the question in which the line law is arbitrary, and Section 6, question 1, asks about that broader possibility. We address its two-leaf branched-ray case, formulated explicitly as AIM Problem 584. All references to [2] concern its arXiv version 3.

Let G have vertices $o, a, b, v_1, v_2, \ldots$ and edges oa, ob, ov_1, v_jv_{j+1} for $j \geq 1$. A proper four-coloring takes values in $\{0, 1, 2, 3\}$ and assigns different colors to the endpoints of every edge almost surely. Graph distance between sets means their minimum vertex distance.

Theorem 1. *There is no pair (X, Y) with the following properties:*

1. *X is a random proper four-coloring of G, and $Y = (Y_j)_{j\in\mathbb{Z}}$ is a random proper four-coloring of the integer line.*
2. *Each process is one-dependent for arbitrary vertex sets at distance greater than one, and each law is invariant under every permutation of $\{0, 1, 2, 3\}$.*
3. *Y is stationary and reflection-invariant.*
4. *For every ordered simple path $(w_1, \ldots, w_m)$ in G, $(X_{w_1}, \ldots, X_{w_m})$ has the law of $(Y_1, \ldots, Y_m)$.*

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No formula for the line law is assumed. Restriction gives the same nonexistence conclusion if either of the two finite arms is lengthened. The proof uses the infinite ray, so it does not assert nonexistence on a fixed finite truncation. It also retains the color and path symmetries in the theorem.

The infinite ray strengthens the usual finite marginal equations. Avoiding one fixed color along its continuation gives a recurrence with a double characteristic root; nonnegativity then bounds any specified extension by half its prefix probability. Short marginal equations express the unknown probabilities using two parameters. All products needed at the chosen finite lengths involve just their square and mixed product. A rational combination of the resulting identities and bounds is a negative constant. Sections 2–3 derive these facts, and Section 4 supplies the exact finite contradiction.

2 Cylinder identities and the infinite tail

Assume such a pair exists. For a color word $w = (w_1, \dots, w_m)$, write

$$p(w) = \mathbb{P}(Y_1 = w_1, \dots, Y_m = w_m), \qquad p(\varnothing) = 1.$$

For $n \geq 0$, write

$$s(A, B, O, r_1, \dots, r_n) = \mathbb{P}(X_a = A, X_b = B, X_o = O, X_{v_i} = r_i \ (1 \leq i \leq n)).$$

These are probabilities of specified colors, not masses of color orbits. Improper words have probability zero. The path condition gives $s(A, B, O) = p(A, O, B)$.

Color relabeling preserves both types of probability. Stationarity and reflection also give $p(w) = p(w^{\mathrm{rev}})$. To name a color pattern, relabel its first distinct color as 0, its next distinct color as 1, and so on. For a line word choose the lexicographically smaller relabeling of the two orientations. For a branched word only relabel colors: the leaf positions a, b remain ordered. When several colors in a sum give the same named pattern, their multiplicities are retained.

Every specified one-site probability is $1/4$, and every proper specified pair has probability $1/12$. For proper line triples,

$$p(i, j, i) = \frac{1}{48}, \qquad p(i, j, k) = \frac{1}{32} \quad (i \neq k). \tag{1}$$

Indeed the endpoints are independent uniforms. For equal endpoints there are three possible middle colors, and for unequal endpoints there are two; permutations fixing the endpoints make the allowed choices equiprobable. The root neighbors a, b, v_1 are likewise jointly independent uniforms. If their specified colors contain d distinct values, then for any allowed root color

$$s(A, B, O, r_1) = \frac{1}{64(4 - d)}, \qquad d = |\{A, B, r_1\}|. \tag{2}$$

Joint independence here follows by applying the arbitrary-set independence condition successively to a singleton and the remaining set. Permutations fixing all occupied neighbor colors act transitively on the $4 - d$ root choices.

Finite additivity and one-dependence give the following identities. Every sum is over the four actual colors $c \in \{0, 1, 2, 3\}$; an improper summand is zero. For line words,

$$\sum_c p(uc) = p(u), \tag{3}$$

$$\sum_c p(ucv) = p(u)p(v) \quad (u, v \neq \varnothing). \tag{4}$$

In (4), removing the intervening site separates the two specified blocks by distance two. Put $r = (r_1, \dots, r_n)$, $n \geq 1$. The corresponding branched identities are

$$\sum_c s(A, B, O, r_1, \dots, r_{n-1}, c) = s(A, B, O, r_1, \dots, r_{n-1}), \tag{5}$$

$$\sum_c s(A, c, O, r) = p(A, O, r), \qquad \sum_c s(c, B, O, r) = p(B, O, r), \tag{6}$$

$$\sum_c s(A, B, c, r) = \frac{1}{16} p(r). \tag{7}$$

The pair of leaves is independent of the entire ray block after removing the root, which proves (7). Finally, for $0 \leq i < n$,

$$\sum_c s(A, B, O, r_1, \dots, r_i, c, r_{i+2}, \dots, r_n) = s(A, B, O, r_1, \dots, r_i)\, p(r_{i+2}, \dots, r_n). \tag{8}$$

Deleting v_{i+1} separates the two indicated vertex sets by distance two. The empty suffix in this formula has probability one.

Lemma 2. *Let E specify the colors on a finite connected line prefix or a branched prefix, with an infinite ray continuing beyond its last vertex. For each color c,*

$$\mathbb{P}(E, \textit{next vertex has color } c) \leq \frac{1}{2}\mathbb{P}(E). \tag{9}$$

Proof. Let a_n be the probability of E followed by n vertices avoiding c. Set $P = \mathbb{P}(E)$ and $Q = \mathbb{P}(E, \text{next vertex has color } c)$. Then $a_0 = P$ and $a_1 = P - Q$. For $n \geq 2$, subtract from a_{n-1} the event that the final vertex has color c. On that event its predecessor automatically avoids c, by properness. Dropping the predecessor from the restrictions leaves the final vertex at distance at least two from all restricted earlier vertices. Its color is an independent uniform, so

$$a_n = a_{n-1} - \tfrac{1}{4} a_{n-2}.$$

The initial values and induction give

$$a_n = 2^{-n}\big[(n+1)P - 2nQ\big].$$

If $P - 2Q < 0$, the bracket is negative for a sufficiently large integer n. This contradicts $a_n \geq 0$ and proves the bound, also when $P = 0$. □

3 Short cylinders and their products

Define

$$\alpha = 48 p(0, 1, 0, 1), \qquad \gamma = 972 s(0, 1, 2, 1, 3, 1). \tag{10}$$

Consider all named proper line words of lengths one through five, and branched words with one through three ray vertices. There are 18 line and 60 branched probabilities. Apply every identity (3)–(8) within these ranges, together with the one-site value and (10). In every product at least one factor has the fixed value given above. This is therefore a rational linear system in the 78 probabilities, with α, γ on its right side.

Lemma 3 (Exact short-cylinder calculation)**.** *The system just described has rank 78 in its probability columns. Its solution is affine in* $(1, \alpha, \gamma)$*, and each line probability is independent of* γ*. In particular,*

$$p(0,1,0,1,0) = \frac{\alpha}{48} - \frac{1}{288}, \tag{11}$$

$$s(0,0,1,2,1,3) = \frac{1}{1152}, \qquad s(0,0,1,2,1) = \frac{1}{256} - \frac{\alpha}{96}, \tag{12}$$

$$s(0,0,1,2,1,2) = \frac{\gamma}{972} - \frac{1}{1152}, \tag{13}$$

$$s(0,1,2,0,2,0) = \frac{1}{768} - \frac{\gamma}{972}. \tag{14}$$

Proof. This is a finite exact linear calculation. In `src/probability_rules.py`, the function `short_affine_table()` enumerates the specified patterns and constructs the equations from the displayed identities. Sparse Gaussian elimination normalizes a nonzero pivot in each probability column, subtracts its rational multiples, and checks consistency of every redundant row. Back substitution produces all 78 affine expressions. Substitution checks every original equation, including rows not chosen as pivots. Thus the expressions are the unique possible values for a hypothetical coloring, rather than one selected solution of an underdetermined system. The supplied verifier repeats this calculation using exact fractions; the five entries above and the vanishing line coefficients of γ can also be read from its returned table. □

Nonnegativity in (11), (13), and (14), and Lemma 2 applied to the prefix and extension in (12), give

$$\frac{1}{6} \le \alpha \le \frac{5}{24}, \qquad \frac{27}{32} \le \gamma \le \frac{81}{64}. \tag{15}$$

For the upper bound on α, the inequality is explicitly $1/1152 \le 1/512 - \alpha/192$.

We now retain all line words of length at most ten and all branched words with at most eight ray vertices. Set the actual moments

$$t = \alpha^2, \qquad h = \alpha\gamma. \tag{16}$$

Every product in (4) or (8) becomes affine in the probability variables and α, γ, t, h. If either factor is fixed, this is immediate. Otherwise a line deletion has two factors of length at least four and total length at most nine. Each thus has length four or five, and Lemma 3 makes both affine in α alone. For a ray deletion, a non-fixed branched prefix has at least two ray vertices and a non-fixed line suffix has at least four vertices. The one deleted site and the limit of eight ray vertices leave the prefix with at most three ray vertices and the suffix with at most five vertices. Both factors are in the short table, and the suffix is independent of γ.

Explicitly, for rational coefficients with $ge = 0$,

$$\begin{aligned}(c + a\alpha + g\gamma)(d + b\alpha + e\gamma) = {} & cd + (ad + cb)\alpha + (gd + ce)\gamma \\ & + ab\,t + (ae + gb)h.\end{aligned}$$

This is an identity at the actual moments (16). In particular, no assertion that arbitrary values of t, h extend to a probability law is needed.

4 An exact rational contradiction

For a line word of length m, specify its odd-position vertices. They are jointly independent uniforms and their event contains the full word event. For a branched word with n ray vertices, use both

leaves and $v_1, v_3, v_5, \ldots$. These vertices are also mutually at distance at least two. Therefore

$$0 \le p(w) \le 4^{-\lceil m/2 \rceil}, \qquad 0 \le s(A, B, O, r_1, \ldots, r_n) \le 4^{-(2+\lceil n/2 \rceil)}. \tag{17}$$

Repeated specified colors do not affect this independence argument. Equations (15) and (16) also give

$$0 \le \alpha \le \frac{5}{24}, \quad 0 \le \gamma \le \frac{81}{64}, \quad 0 \le t \le \frac{25}{576}, \quad 0 \le h \le \frac{135}{512}. \tag{18}$$

To state the finite calculation, let F_i denote a left side minus right side of a marginal or deletion identity, the one-site identity, or a parameter definition. Products are evaluated as above. Let H_j be one of the following affine expressions:

- $\mathbb{P}(\text{prefix})/2 - \mathbb{P}(\text{specified one-step extension})$;
- $t - 2c\alpha + c^2$, with $c = 5/24$;
- a probability or one of α, γ, t, h;
- its upper bound in (17) or (18), minus that quantity.

Every F_i is zero and every H_j is nonnegative for the hypothetical pair. For the second expression this follows from $(\alpha - c)^2 \ge 0$.

Proposition 4 (Exact certificate)**.** *The supplied file* `certificate/farkas.json` *specifies rational numbers λ_i and nonnegative rational numbers μ_j such that*

$$\sum_{i=1}^{8245} \lambda_i F_i + \sum_{j=1}^{7244} \mu_j H_j = -D, \qquad D = \frac{46631735540736216506687 01}{22960145839712215028989952 0} > \frac{1}{50}. \tag{19}$$

All probability and moment coefficients on the left cancel exactly.

Proof. The JSON entries identify each equation by its geometry, color word and deleted position, rather than by a stored matrix row. Each nonnegative expression identifies its rule and its word or variable. Every multiplier is a literal rational number. The file has SHA-256

`c64ca628b2ea2b324a840827ef7571b9d68f2a8fd4039c7c0baab43581bf4f67`

The function `verify()` in `src/verify_certificate.py` performs the following finite operations:

1. Derive the short affine table by the rational elimination in Lemma 3.
2. Reconstruct each F_i from (3)–(8), the one-site value or (10); expand its products using (16).
3. Reconstruct each H_j from its stated rule and multiply by its recorded weight, rejecting a negative μ_j.
4. Add every coefficient and constant with exact fractions. Check that all 8,298 probability and moment coefficients are zero, that the negative constant equals the displayed $-D$, and that $D > 1/50$.

The nonnegative expressions consist of two tail inequalities, one square, 3,647 lower bounds and 3,594 upper bounds. In particular any small residual from finding the multipliers is canceled by additional valid bounds in the exact identity; it is not an arithmetic tolerance. No optimizer or floating point operation enters this replay. These operations verify (19) from the supplied finite rational data. □

Proof of Theorem 1. The assumed pair supplies its actual finite-cylinder probabilities and moments. The preceding arguments show that every F_i vanishes and every H_j is nonnegative at these values. Since every μ_j is nonnegative, the left side of (19) is nonnegative. Its right side is strictly negative by Proposition 4, a contradiction. □

Reproduction. The supplied code and tests accompany the mathematical certificate. From the supplied directory, run

```
python3 src/verify_certificate.py certificate/farkas.json
python3 -m unittest discover -s test -v
```

Only the Python standard library is required. The first command repeats the proof-critical finite calculation; the tests also reject corrupted certificates and compare short identities with separate probability controls. The decimal approximation printed for convenience is not used in a verification decision. These checks accompany the analytic derivations above; they do not assert independent human or formal verification.

AI-use statement. OpenAI Codex was used for literature retrieval, mathematical exploration, the derivation and checking of the proof, certificate and verifier development, and manuscript preparation. The finite verification is reproducible with the supplied rational certificate and code. Independent mathematical review is requested; no human or Lean certification is claimed.

References

[1] A. E. Holroyd and T. M. Liggett, *Finitely dependent coloring*, arXiv:1403.2448v4 (2015). arXiv source.

[2] T. M. Liggett and W. Tang, *One-dependent colorings of the star graph*, arXiv:1804.06877v3 (2023). arXiv source.